

Bio-Based Anti-Biofoulant with Copper(II) as Intramolecular Catalyst for Radicals Formation

Jinbo Chen, Guangming Lu, Yu Hao, Yichao Lu, Liang-Feng Huang, and Yajie Zhang*

Highly efficient and eco-friendly antimicrobial agents are crucial to the development of future marine antifouling coatings. A high efficient and low bio-toxicity bio-based anti-biofoulant, 2,5-diformylfuran dioxime chelating with copper ions (E-DFFD-Cu), is evaluated. Its minimal bactericidal concentration and the concentration for 50% of maximal effect (EC₅₀) are 4.95 and 7.03 $\mu\text{mol L}^{-1}$, respectively. Excellent anti-biofouling performance in marine environment in situ can be kept at least 90 days and with maximum release rate of copper ion of 0.01 $\mu\text{mol cm}^{-2} \text{d}^{-1}$. Alkyl radical (R·) and hydrogen radical (H·) derived from E-DFFD are enhanced via chelating copper(II) catalysis, which attracts and disrupts microbe cell membrane and pili between intimate cells, resulting in excellent anti-biofouling effect. This provides a new perspective for developing anti-biofoulant from anti-biofouling mechanism in the future.

1. Introduction

Marine biofouling is an important part of marine economic loss, and excessive fouling accumulation can cause potential risks and negative consequences to hull ships, offshore oil platforms, coastal facilities and aquaculture units, etc.^[1–4] The most important method to prevent biofouling is using antifouling coatings with highly effective antifoulants.^[5–7] In the early development of antifouling coatings, toxic compounds (e.g., mercury, lead, arsenic, organotin, etc.) were used as the functional components.^[8] For example, the early tributyltin (TBT) as an effective antifoulant has been widely used since early 1970s,

but was completely banned by the International Maritime Organization in 2002 due to its ability to form enrichment effect in organisms through the food chain, affecting food safety.^[9,10]

At present, copper oxide, organic copper or a combination of several copper-based antifoulants are the mainstream anti-biofoulants used in antifouling coatings.^[11,12] Copper oxide is the most widely used antifouling agent due to its low cost and broad-spectrum bactericidal effect.^[13] However, it has become evident that copper is highly toxic to marine invertebrates in the marine environment, especially to their larvae. Some research have shown that CuCl₂ at a concentration of 0.38 $\mu\text{mol L}^{-1}$ in the environment can cause complete

arrest of oyster (*Crassostrea gigas*) larvae, while the semi-lethal concentration (LC₅₀) and absolute lethal concentration (LC₁₀₀) for Virginia oyster (*Crassostrea virginica*) embryos are 0.78 and 0.98 $\mu\text{mol L}^{-1}$, respectively.^[14] For crustaceans, copper was able to kill 50% of different crustacean species after 48 h of exposure to the environment, and juveniles were more sensitive than adults. According to statistics, more than 3000 metric tons of copper are currently imported from the coating into seawater each year.^[15] Especially in some bays or closed water bodies where water exchange is poor, copper-containing antifouling coating has a significant effect on the copper content of the water body.^[13] In addition to copper-containing antifoulants, chlorothalonil, Irgarol 1051, Sea-Nine 211, diuron, zinc dioxane, etc., are commonly used as eco-friendly antifoulants, but these antifoulants are still required to be combined with cuprous oxide or other copper-containing antifoulants because of their weak inhibitory effect on large fouling organisms.^[16–18] Therefore, the development of efficient and eco-friendly antifoulants that can replace and significantly reduce of copper-containing antifoulants is crucial for antifouling coatings.

Oxime compounds are one of the most important precursors for the preparation of biologically active antibacterial,^[19] and copper ions can initiate oxime to produce iminoxyl radicals via the homolytic cleavage of O–H bonds,^[20] which have highly reactive free radicals in exerting effective antibacterial activity in vitro against most Gram-negative organisms.^[21] In the early experiment, some of the antifouling coatings were accidentally contaminated by 2,5-diformylfuran dioxime (DFFD) and its metal chelates. These contaminated antifouling coatings were free from any microorganisms, while some control coatings containing antimicrobial agents showed different degree of microbial growth. Additionally, the solubility of DFFD is

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very low in aqueous solution, and DFFD were prepared from the renewable and sustainable hexose. All of these imply that DFFD from renewable sources has the potential to be used as an efficiently antifoulant for antifouling coatings. However, there are no any reports on DFFD in anti-fouling coating. Thus, we tentatively investigated that *E*-isomer of DFFD and its copper(II) complexes (*E*-DFFD-Cu) as anti-biofoulants in marine anti-fouling coatings.

2. Results and Discussion

2.1. Structure Characterization of *E*-DFFD-Cu

Bio-based *E*-DFFD (Figures S2–S4, Supporting Information) was prepared according to the literature,^[22] and *E*-DFFD-Cu

complex were prepared with cupric chloride at room temperature (Figure 1a,b).

The scanning electron microscope (SEM) images and EDS mappings demonstrate that copper is uniformly distributed in the complex, suggesting that copper ions have been successfully coordinated to the ligand (Figure 1c). The elemental analysis showed that the copper(II) ion is possibly coordinated by four nitrogen atoms of *E*-DFFD (Figure 1b). Three differences of infrared spectra (FTIR) of *E*-DFFD and *E*-DFFD-Cu further identified. First, asymmetric oxime OH...O stretching vibrations appear at 3446 cm⁻¹ and 3355 cm⁻¹ (Figure 1d), respectively, indicating the formation of intermolecular hydrogen bonds between –OH groups in the oxime ligands.^[23] Second, the typical (C=N oxime) stretching vibration at 1637 cm⁻¹ shifts to 1598 cm⁻¹ upon complexation (Figure 1e), implying that nitrogen atoms are coordinated with the metal ions.^[24]

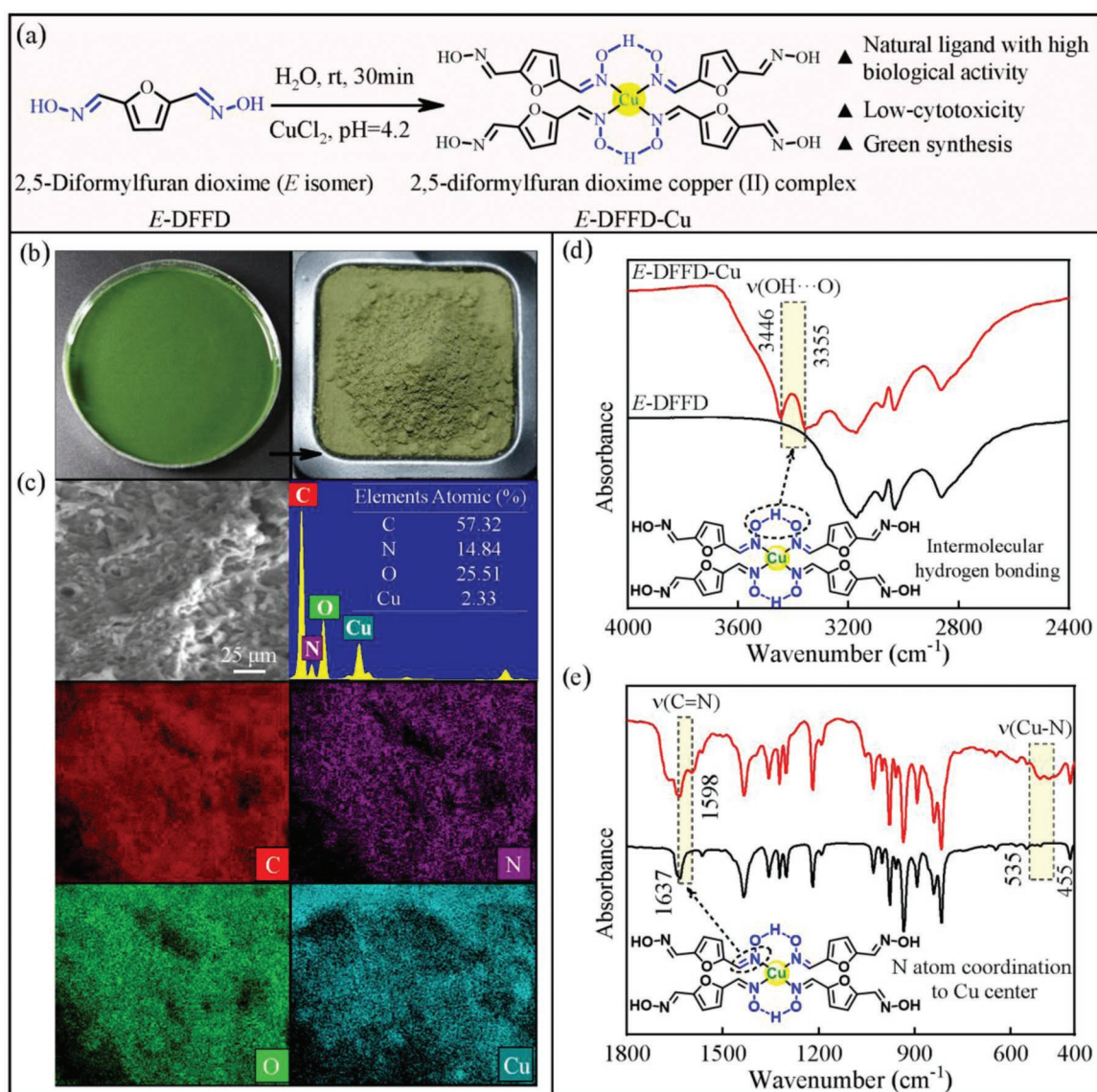


Figure 1. a) Synthetic scheme of the *E*-isomer of 2,5-diformylfuran dioxime (*E*-DFFD) and its copper(II) complex (*E*-DFFD-Cu). b) *E*-DFFD-Cu solution and powders. c) SEM images and EDS spectra of *E*-DFFD-Cu. FTIR spectra of *E*-DFFD and *E*-DFFD-Cu in the range of d) 4000–2400 cm⁻¹ and e) 1800–400 cm⁻¹, respectively.

Moreover, a new peak below 550 cm^{-1} in the complex is observed assigned to Cu–N stretching vibration (Figure 1e).^[25] These results clearly illustrated the coordination of the ligands through the nitrogen atoms of the C=N group.

2.2. Bactericidal Activity and Anti-Microalgae Adhesion Performance

According to the theory of marine environmental biofouling occurrence, the order of biofouling occurrence is bacteria – microalgae – barnacles – macroalgae. Therefore, the antibacterial performance and microalgae antiadhesion effect were first tested.

Escherichia coli (ATCC 25922) and *Vibrio parahaemolyticus* (ATCC 33844), two typical Gram-negative bacterial, were firstly choose to test their MBC of *E*-DFFD and *E*-DFFD-Cu via a traditional shake flask method.^[26,27] The results indicated that both were effective in killing bacteria while less effective in killing *E. coli* (Figure 2a,b) than *V. parahaemolyticus* (Figure 2c,d).

For *E*-DFFD, the percentage of *E. coli* killed in 24 h was 85.92% at $4.95\text{ }\mu\text{mol L}^{-1}$, which was further enhanced to 98.74% when the concentration increased to $19.80\text{ }\mu\text{mol L}^{-1}$. *E*-DFFD-Cu demonstrated remarkable bactericidal activity and killed $\approx 100\%$ *E. coli* even at $4.95\text{ }\mu\text{mol L}^{-1}$ after 12 h (Figure 3a), which was much lower than the minimum inhibitory concentrations (MIC) of other antibacterial agents (Table S1, Supporting Information).^[28–33] Compared with *E*-DFFD ($\text{MBC}_{24} > 19.80\text{ }\mu\text{mol L}^{-1}$)

and CuCl_2 ($\text{MBC}_{24} \approx 105.37\text{ }\mu\text{mol L}^{-1}$), excellent killing bactericidal performance of *E*-DFFD-Cu indicated that copper ions coordinated in the complex possibly play a crucial role in bactericidal activity. Moreover, the elemental analysis also demonstrated that the copper concentration was only $5.72\text{ }\mu\text{mol L}^{-1}$ in solution (Figure 1c), significantly lower than the CuCl_2 control ($12.57\text{ }\mu\text{mol L}^{-1}$). This further suggests that the oxime group chelated copper ions have a possible synergistic bactericidal effect.

Marine microalgae are also the main biofouling organism influencing biofilm formation after the initial fouling stage in the marine environment.^[34,35] The effect of these antifouling coatings on the algae attachment of representative *Phaeodactylum tricornutum* and *Nitzschia closterium* forma *minutissima* was further explored to confirm the antifouling effect.

As shown in Figure 3, large amounts of microalgae colonized on the coating surface in the absence of *E*-DFFD and *E*-DFFD-Cu. The adherence area of both *P. tricornutum* and *N. closterium* f. *minutissima* gradually reduced with increasing concentration of *E*-DFFD. However, there were still 0.94% and 0.92% of the coating surface for *P. tricornutum* and *N. closterium* f. *minutissima*, respectively, treated with 10 wt% *E*-DFFD at 24 h (Figure 3a). This indicated that *E*-DFFD cannot completely prevent both microalgae adherence at low concentration. Similar to the *E*-DFFD-Cu performance in killing bacterial, the adherence areas of *P. tricornutum* and *N. closterium* f. *minutissima* incubated with *E*-DFFD-Cu coating were significantly lower than that of *E*-DFFD, showing almost 100% killing both microalgae

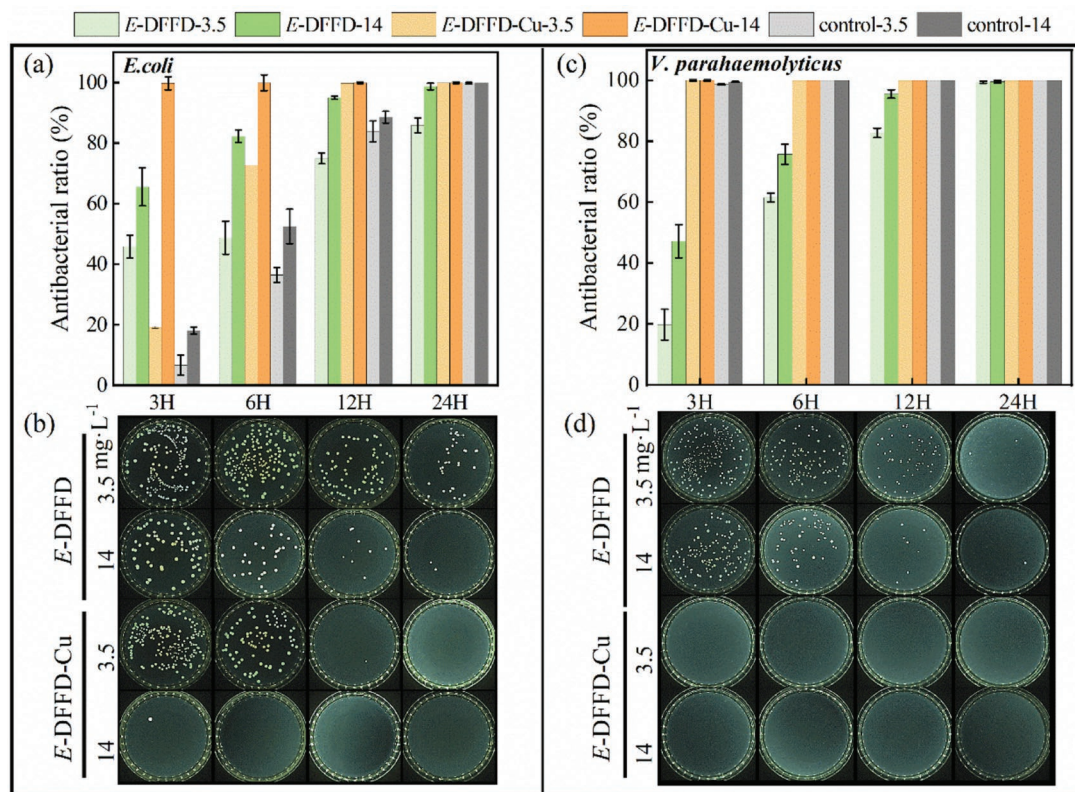


Figure 2. The bactericidal performance of *E*-DFFD and *E*-DFFD-Cu toward a,b) *E. coli* (ATCC 25922) and c,d) *V. parahaemolyticus* (ATCC 33844), respectively.

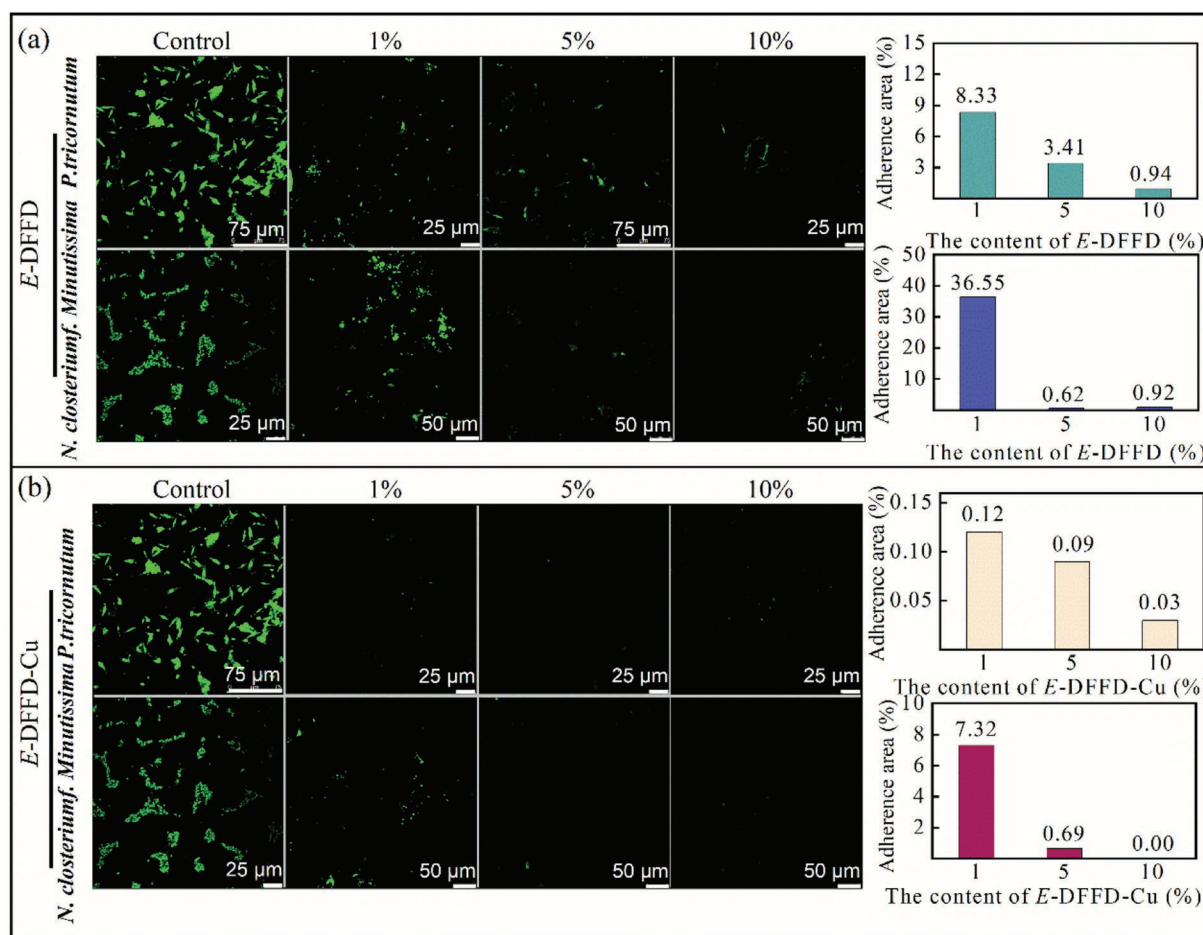


Figure 3. The fluorescence microscopies of *P. tricornutum* and *N. closterium f. minutissima* adherence to the coating surface with different contents of a) *E-DFFD* and b) *E-DFFD-Cu*.

with 10 wt%, and almost no microalgae were observed on the *E-DFFD-Cu* coating surface above 5 wt% (Figure 3b).

2.3. Antifouling Testing in Marine Environments In Situ

Based on the above excellent antifouling performance of *E-DFFD* and *E-DFFD-Cu*, the antifouling ability was further investigated in the marine environment in situ. As shown in Figure 4b, the control panels coating in 304 stainless steel and pure resins showed significant visible biofouling (e.g., barnacles and lime worms) after 30 days, while all experimental panels with varying loading contents from 1 to 10 wt% showed no significant fouling on their surfaces. This performance remained unchanged until 60 days of exposure (Figure 4c), though small amounts of algae adherence are observed on the panels coated with 1 and 5 wt% *E-DFFD*. After 90 days of immersion (Figure 4d), excellent antifouling performance was still observed for the 10 wt% *E-DFFD-Cu* coatings with small amounts of algae attachment. Compared with the control experiments, this illustrated that both *E-DFFD* and *E-DFFD-Cu* have anti-biofouling effect, and *E-DFFD-Cu* was better, indicating its potential for anti-biofouling.

Since our marine experiment site is in the East China Sea, located at the mouth of the Yangtze and Qiantang rivers, a large amount of sediment was brought in. Therefore, the amount of sediment was relatively large throughout the sea, and the off-shore seawater appeared yellow. Thus, the surface of the 90 day coating still remained sediment after a simple rinse and appeared yellow. However, a large number of barnacles grew on the control resin. According to the barnacle attachment mechanism, it should be first to grow bacteria, then microalgae, and then barnacles. The coating containing 10% *E-DFFD-Cu* is almost no barnacles, much less green microalgae, which indicated that *E-DFFD-Cu* has excellent antifouling effect, and the fouling was mainly caused by sediment.

2.4. Possible Antimicrobes Mechanism

As a new anti-biofoulant, there is still no any knowledge about its anti-biofouling mechanism. As shown in Figure 5a,b, both *E-DFFD* and *E-DFFD-Cu* damaged the integrity of the *E. coli* cell membrane, and the disruption extent of *E-DFFD-Cu* was greater than that of *E-DFFD*. Both bacteria cell were entirely disrupted by *E-DFFD-Cu* with $4.95 \mu\text{mol L}^{-1}$ at 24 h. As

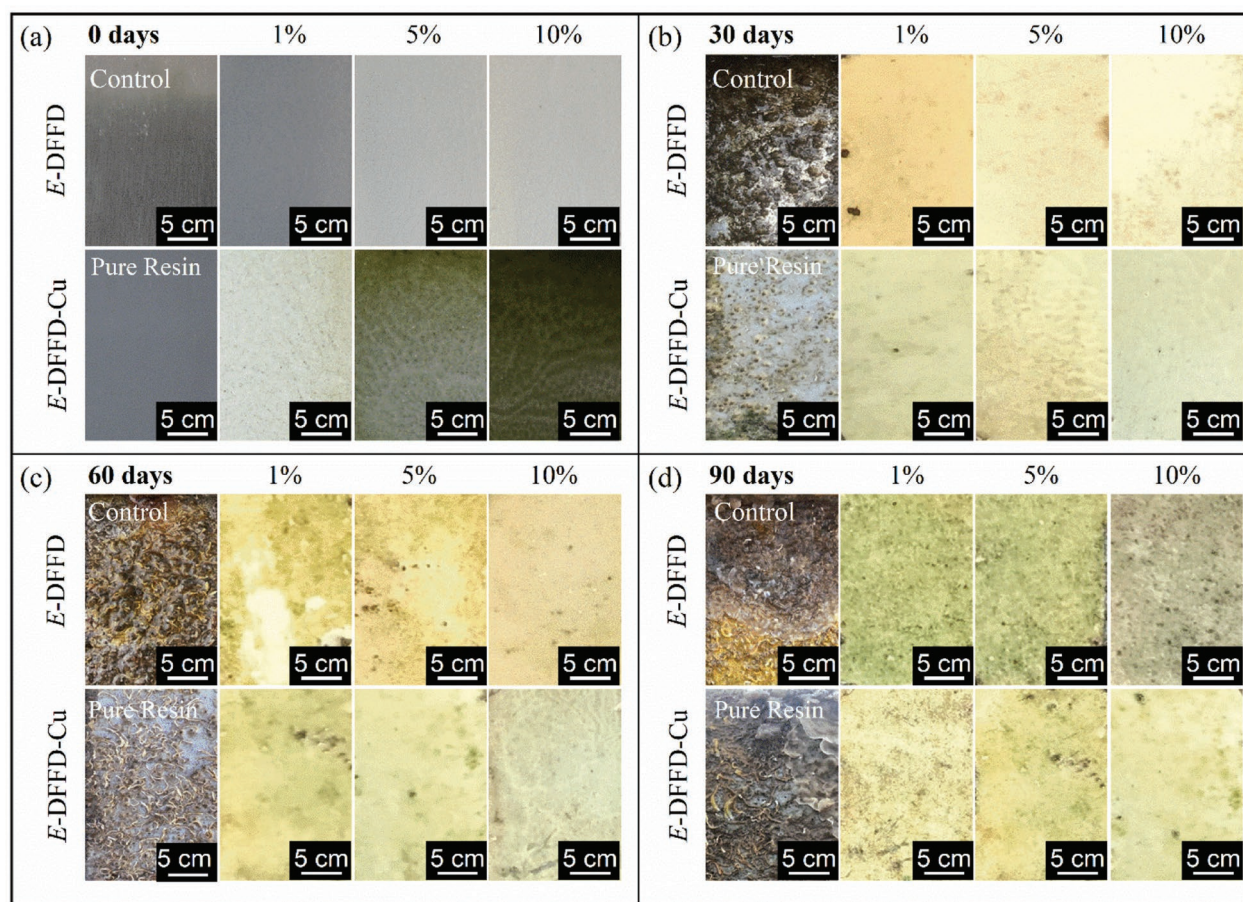


Figure 4. Antifouling evaluations of *E-DFFD* and *E-DFFD-Cu* coatings with different contents in real marine environment in situ for a) 0, b) 30, c) 60, and d) 90 days, respectively.

shown in Figure 5c,d, *E. coli* cells were severely contracted and deformed, and cellular exudates or disintegrants remained at the cell periphery, indicating that membrane was completely destroyed. In addition, the conjugative pili of *E. coli*, a passage for horizontal gene transfer to increase resistance to antibacterial or antibiotics, was broken down (Figure S5, Supporting Information), indicating the communication between intimate cells was interrupted.

In term of *E-DFFD-Cu* molecular structure, the killing microbe mechanism of free radical is speculated. Figure 6a presents the EPR spectra of DMPO spin adducts in saline solution containing *E-DFFD* and *E-DFFD-Cu* of $4.95 \mu\text{mol L}^{-1}$. As anticipated, the dissolution of *E-DFFD* in water at room temperature produced a predominant free radical which was identified as a DMPO-H adduct showing a primary triplet ($a_N = 16.32 \text{ G}$) that was further split by two secondary hydrogens ($a_H^\beta = 22.17 \text{ G}$). These hydrogen radicals ($\text{H}\cdot$) are likely produced by homolytic cleavage of O–H bond with the simultaneous formation of the iminoxyl radicals ($\text{C}=\text{N}-\text{O}\cdot$). However, the other peaks cannot be iminoxyl radicals, since the characteristic iminoxyl ^{14}N hyperfine coupling constants, 28–33 G,^[36,37] are far greater than any observed in this study. A theoretical simulation (Figure 6b, and see the simulation methods and raw data in the supplementary information 2–4) using the coupling

constants and the observed line width values nicely reproduce the observed spectrum. The equal nitrogen and hydrogen hyperfine coupling constants ($a_N = 15.67 \text{ G}$, $a_H^\beta = 22.65 \text{ G}$) with an intensity ratio of 1:2:2:1 are characteristic of the alkyl radical species ($\text{R}\cdot$),^[36–38] which may be formed from the delocalization of the unpaired electron between the nitrogen and carbon atoms in the iminoxyl radicals. This unique radical is extremely toxic and mostly probably damage the cell membrane structure, thus presenting a substantial impact on the antimicrobial activity employed in this study. Notably, a significantly enhanced intensity of the same two species with the copper complex solution was observed, indicating that the addition of Cu(II) ions should provide a pathway for *E-DFFD* to produce more free radicals.

Theoretical calculations provide more insight into the formation mechanism of two species free radical. The DFT calculations of iminoxyl radicals indicated an apparent charge transfer from the copper atom at the coordination center to the oxygen atom on the oxime hydroxyl group (Figure 6c). The charge transfer resulted into the homolytic cleavage of the O–H bond more easily compared with *E-DFFD* alone. The bond dissociation energy of the O–H group decreases from 82.56 to 57.19 kcal mol^{-1} under the catalysis of copper ions (Figure 6d). This confirmed the above speculation of a copper ion catalytic effect on

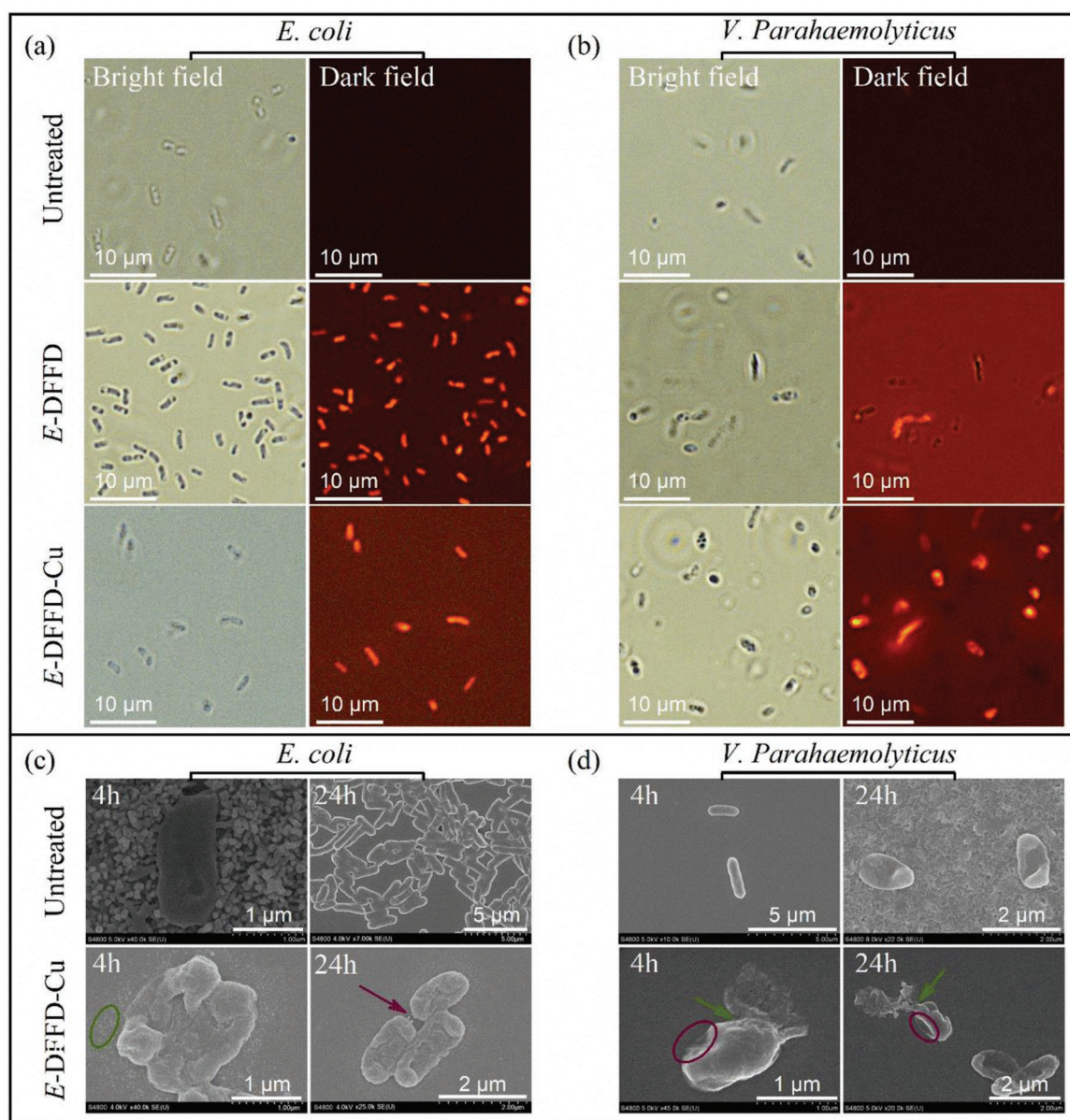


Figure 5. Bright field (left) and dark field (right) images of propidium iodide staining on a) *E. coli* and b) *V. parahaemolyticus* with and without treatment by *E*-DFFD and *E*-DFFD-Cu, respectively, at $4.95 \mu\text{mol L}^{-1}$. SEM images of c) *E. coli* and d) *V. parahaemolyticus* cells with and without treatment by *E*-DFFD-Cu at $4.95 \mu\text{mol L}^{-1}$ (green circle: the cellular exudates or disintegrants; magenta arrow, the broken conjugative pili; green arrow, the broken hole; magenta circle, the retention of the outer membrane).

homolytic cleavage derived from charge transfer. Moreover, the electronic potential difference ($\Delta\phi$) between the N-O oxygen atom and hypo-methyl carbon atom was 0.25 eV (Figure 6e), implying that the excess electrons are prone to be transferred to the carbon atom, facilitating alkyl radical generation. It is well known that the unpaired electron on the iminoxyl radical is easily delocalized from the iminoxyl's oxygen to the sp^2 hybridized carbon^[36] and hence transferred into the furan aromatic ring due to conjugation with the π -system.^[38] This result is further supported by the bond length difference between the C=N and C-N bonds and atomic-charge changes to the carbon atom before and after hydrogen dissociation of the oxime group (Figure 6f). The double bond length of C=N increases by

$0.05 \text{ \AA}$ after hydrogen dissociation, indicating the appearance of a single bond. The atomic charge of the hypo-methyl carbon atom increased from 3.42 e^- to 3.52 e^- , indicating electron accumulation on the carbon atom. The above results suggested that the generation of alkyl radicals from oxime groups is responsible for the remarkable antibacterial and antifouling properties. In addition, copper ions at extremely low concentrations (i.e., $5.72 \mu\text{mol L}^{-1}$) in the complex act as an effective catalyst for the spontaneous formation of these radicals used to kill Gram-negative bacteria.

According to the above results, a possible mechanism was tentatively proposed for the antifouling performance of *E*-DFFD-Cu coating (Figure 7). Iminoxyl and hydrogen radicals are

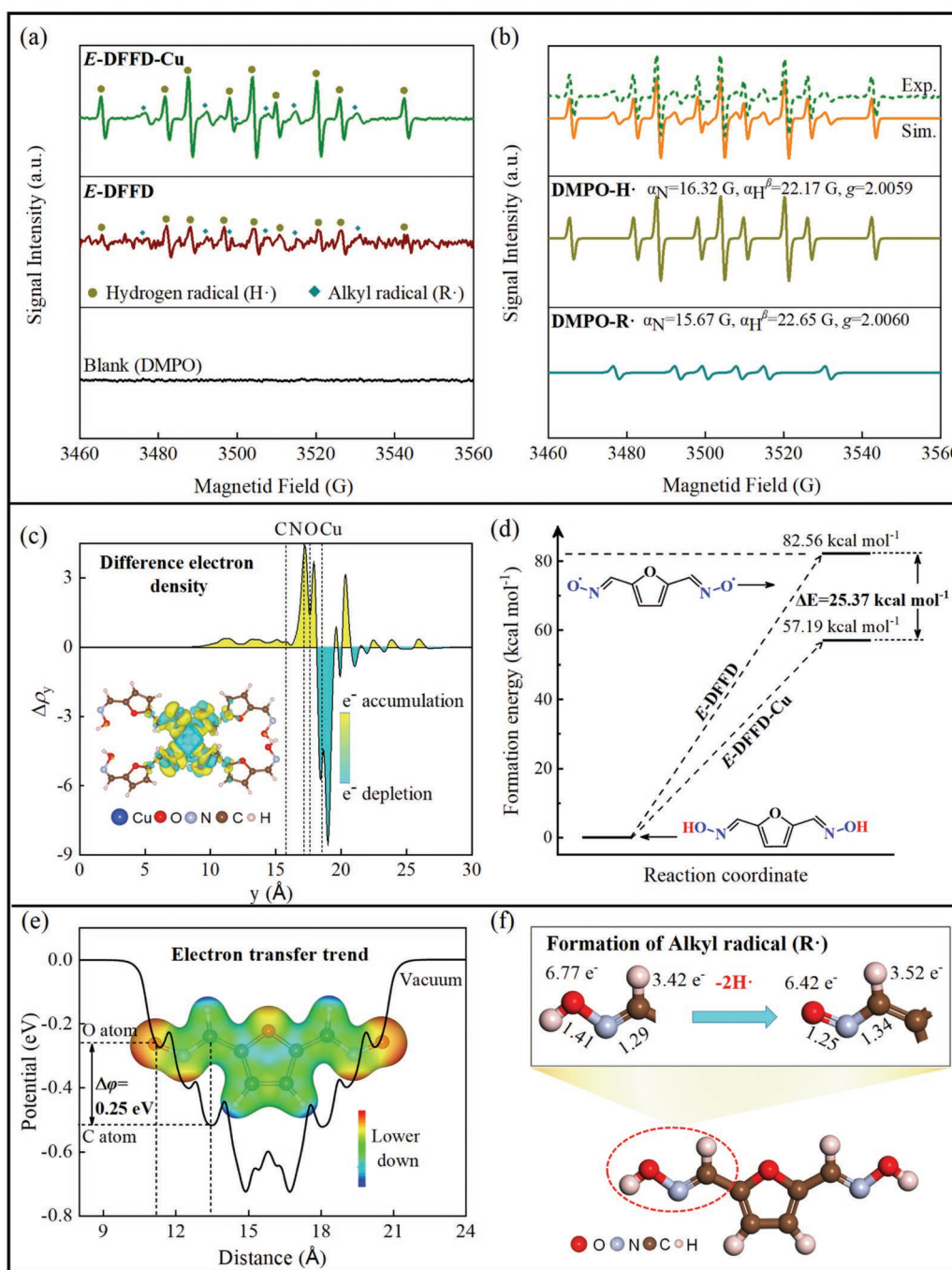


Figure 6. a) EPR spectra of DMPO (1.0 mmol L⁻¹) spin adducts in the phosphate-buffered saline solution (pH 7.4) containing E-DFFD and E-DFFD-Cu (4.95 μmol L⁻¹). b) Simulated spectra of adducts. The dashed and solid lines indicate the characteristic peaks obtained from experimental and simulated results, respectively. c) The distribution of the plane-averaged differential electron density (Δρ_y, where y is the long molecular axis) caused by copper chelation in E-DFFD-Cu. d) Calculated formation energies of the iminoxyl radical from E-DFFD and E-DFFD-Cu. e) Electronic-potential profile along the y-axis of iminoxyl radical with free O atom. f) Bond lengths and atomic charges of the E-DFFD molecule before and after hydrogen dissociation.

continuously and spontaneously produced via homolytic O–H bond cleavage in the E-DFFD compound in an aqueous solution. The resultant iminoxyl radicals were further converted into alkyl radicals through the charge flow action from the oxygen atom of the N–O species to the hypo-methyl carbon atom, driven by the potential difference between these two atom types.

However, the dissociation process of E-DFFD was relatively slow at room temperature without metal involvement or energy input from light or heat. By contrast, the copper ions cooperated in the E-DFFD-Cu complex facilitated oxime O–H bond dissociation process, which reduced the bond dissociation enthalpy of the O–H bonds and enhancing the charge transfer from the copper

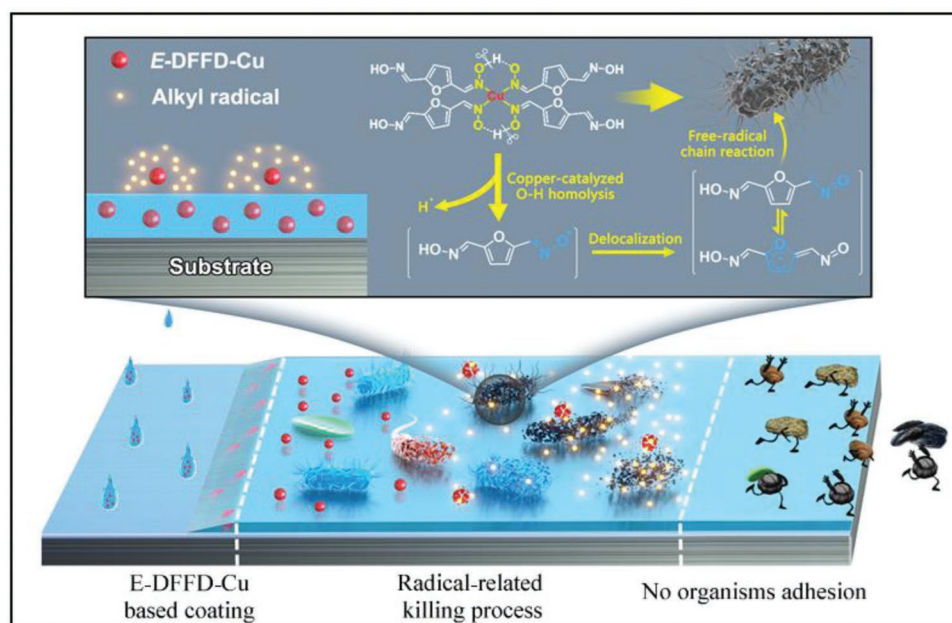


Figure 7. Schematic representation of active radical generation under copper catalysis and antifouling mechanisms of *E*-DFFD and *E*-DFFD-Cu coatings.

center to oxygen atoms on the oxime hydroxyl group, leading to alkyl radical formation. These highly unstable and reactive alkyl radicals further damage the cell membranes of bacteria and microalgae. Therefore, *E*-DFFD-Cu composite coatings with higher concentrations of active alkyl radicals and effectively prevented the formation of infectious bacterial biofilms. Subsequently, the adhesion of main fouling organisms, such as bacterial, algae, moss worm and barnacles, was prevented.

2.5. Release Rate of Copper Ion and Bio-Toxicity of *E*-DFFD-Cu

In order to evaluate the pollution to the marine environment, the release rate of copper ion was further investigated. As shown in **Figure 8a**, the coating with 1 wt% *E*-DFFD-Cu had a low and nearly constant release rate of about $8.5 \times 10^{-4} \mu\text{mol cm}^{-2} \text{d}^{-1}$

throughout the experiments. For the coatings with 5 and 10 wt% *E*-DFFD-Cu, the release rates were relative higher, and none of the coatings presented an excessive release at the early stages. After 16 days of immersion, the release rates of *E*-DFFD-Cu with 5 and 10 wt% gradually decreased with time extension, leveling off at 4.0×10^{-3} and $9.1 \times 10^{-3} \mu\text{mol cm}^{-2} \text{d}^{-1}$, respectively. More importantly, the maximum release rate of copper ions was only $0.01 \mu\text{mol cm}^{-2} \text{d}^{-1}$ with the 10 wt% *E*-DFFD-Cu. This release rate is about 24 times lower than reported for copper-based self-polishing antifouling coatings (normally $0.24 \mu\text{mol cm}^{-2} \text{d}^{-1}$),^[39,40] indicating that *E*-DFFD-Cu coatings can effectively reduce copper contamination risk on marine environment. This is lower than the China standard of ship anti-fouling paints (HJ 2515-2012) of less than $0.393 \mu\text{mol cm}^{-2} \text{d}^{-1}$ and the Canadian standard of $0.629 \mu\text{mol cm}^{-2} \text{d}^{-1}$ for all copper-containing antifouling coatings.

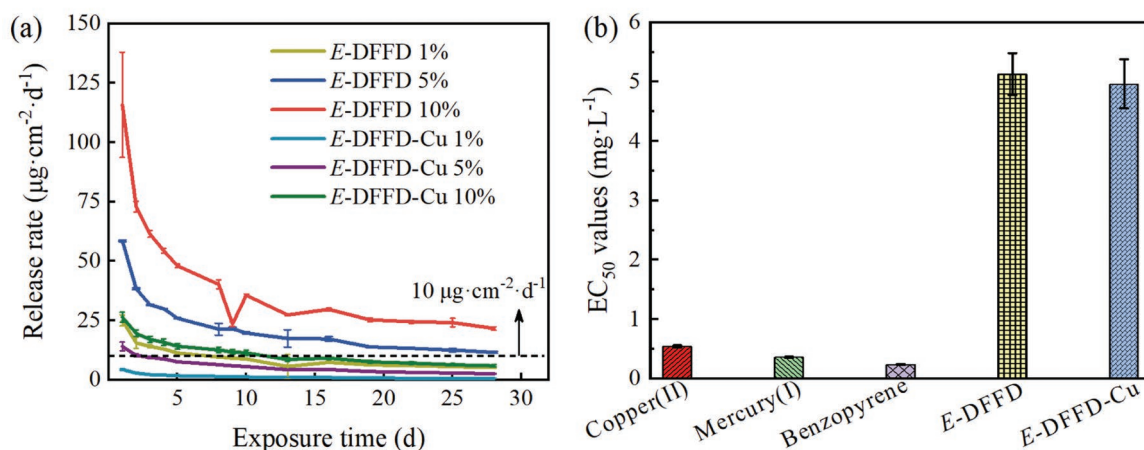


Figure 8. a) The average release rate of *E*-DFFD and *E*-DFFD-Cu from coatings with different contents. b) The EC₅₀ of *E*-DFFD and *E*-DFFD-Cu using copper(II), mercury(I), and benzopyrene as positive control agents.

The low release rate can be attributed to four possible factors. First, the solubility of *E*-DFFD-Cu itself in the water environment is less than 1000 ppm. Second, the content of the added content of *E*-DFFD-Cu was 10 wt%, which translated into a copper ion concentration of only 141.42 $\mu\text{mol L}^{-1}$, resulting in a low release rate. Third, *E*-DFFD-Cu is an organic compound, which chemically interacts with coating resin to some extent, which will also slow down the overall penetration from the inside to the outside. In addition, its chelated copper ions with *E*-DFFD are not simply mixed with the coating resin, with a certain chemical interaction, also limiting the copper ions from the inside to the outside penetration and release into seawater. Fourth, the outside *E*-DFFD-Cu cured in the coating resin dissolved in seawater, then the inside *E*-DFFD-Cu of the coating resin gradually penetrate outward, which resulted in a slower release rate of dissolution. The combination of four factors above leads to the low rate of copper ion release.

Bio-toxicity is a key indicator for the development of eco-friendly anti-biofoulant, and we thus focused on comparing *E*-DFFD-Cu with three typical substances for biological toxicity evaluation, Cu (II) ($\text{EC}_{50} = 8.50 \mu\text{mol L}^{-1}$), mercury(I) ($\text{EC}_{50} = 1.79 \mu\text{mol L}^{-1}$), and benzopyrene ($\text{EC}_{50} = 0.95 \mu\text{mol L}^{-1}$). EC_{50} of *E*-DFFD and *E*-DFFD-Cu to *V. fischeri* were 33.3 and 703 $\mu\text{mol L}^{-1}$, respectively. This demonstrated that *E*-DFFD-Cu was at least 10 times less bio-toxicity compared to three controls (Figure 8b). Therefore, *E*-DFFD-Cu is low bio-toxicity and can be considered as an eco-friendly anti-biofoulant.

3. Conclusion

A new biobased anti-biofoulant, *E*-DFFD-Cu, was developed and tested its performance in anti-biofouling coating. *E*-DFFD-Cu can effectively kill Gram-negative bacterial and prevented the adhesion of microalgae, which is the initial critical aspect of preventing biofouling. The marine anti-biofouling excellent performance in situ, low bio-toxicity and slow copper ion release indicated that *E*-DFFD-Cu can be considered as a potential alternative of the current antifoulants. Subsequently, we tentatively proposed a possible radical mechanism of the anti-biofouling catalyzed by chelated copper(II) ion, which provides a new perspective for developing anti-biofoulant from anti-biofouling mechanism in the future.

4. Experimental Section

Materials: 2,5-Diformylfuran (DFF, 99%) was gifted by Ningbo Biomass and Biotechnology Co., Ltd (NBB). *N,N*-Dimethylformamide (DMF, 99.5%), methanol (MeOH, 99.5%), hydroxylamine hydrochloride ($\text{NH}_2\text{OH HCl}$, 99.0%) and sodium acetate (NaOAc , 99.0%) were purchased from Aladdin Chemistry Co. Ltd. (Shanghai, China). Copper chloride dihydrate ($\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$, 99.99%) was obtained from the Sinopharm Chemical Reagent Beijing Co., Ltd (Shanghai, China). Dimethyl sulfoxide- d_6 ($\text{DMSO-}d_6$, 99.8 atom% D) and 5,5-dimethylpyrrolidine-*N*-oxide (DMPO, 98%) were supplied by Sigma-Aldrich chemical Reagent Co., Ltd (Shanghai, China). All the reagents were of reagent grade and used without further purification.

Synthesis of *E*-DFFD-Based Copper(II) Complex: 50 mL solution containing *E*-DFFD (11.54 mg, 10 mmol) was titrated with 100 μL increments of 4.0 mol L^{-1} freshly prepared NaOH dosed at 1 min

intervals (pH 12.03). 50 mL $\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$ (1.705 g, 10 mmol) solution was then added into the prepared *E*-DFFD solution. The mixture continued stirring for another 30 min at room temperature, then diluted aqueous HCl (15 wt%) was added slowly into the mixture until the pH 4.2. The dark green precipitate formed was filtered off (Figure S1, Supporting Information), washed with distilled water ($3 \times 30 \text{ mL}$) and then freeze dried at -20°C overnight. The copper complex was collected by solvent evaporation as dark green power (1.55 g, 91% isolated yield with respect to *E*-DFFD).

Antibacterial Tests: Typical Gram-negative bacteria, *E. coli* (ATCC 25922) and *V. Parahaemolyticus* (ATCC 33844), were employed as model bacterial to investigate the antibacterial activity of *E*-DFFD and *E*-DFFD-Cu through shake flask method (ASTM E2149-01).^[41] Series of diluted *E*-DFFD-Cu solution in the range of 2.475–19.80 $\mu\text{mol L}^{-1}$ were prepared by twofold dilution method. *E. coli* (ATCC 25922) were then added to the diluted *E*-DFFD-Cu solution to a final bacterial concentration of $1 \times 10^5 \text{ CFU mL}^{-1}$ and incubated at 37°C for 24 and 48 h, and then transferred to agar plate and kept incubating for 24 h. If no bacterial was observed in the agar plates, the corresponding relative minimum concentration of *E*-DFFD-Cu was considered as MBC. The positive control groups contain *E. coli* without *E*-DFFD-Cu. The negative control groups were devoid of both bacterial inoculum and *E*-DFFD-Cu. 4.95 $\mu\text{mol L}^{-1}$ was finally determined as the MBC for *E. coli*.

Anti-Microalgae Tests: Two microalgae model species (*Phaeodactylum tricornutum* NMBgu001 and *Nitzschia closterium* f provided by the Microalgae Collection Center at Ningbo University) were cultured under 1.0 mW cm^{-2} illumination (12/12 dark/light cycle model) at 20°C in F/2 medium. Prior to experiments, these marine algal cell suspensions were diluted to 10^8 algal cells mL^{-1} using F/2 medium. Then 1.0 mL of the microalgae suspension was transferred to the 24-well plates containing the agents to test. The plates were incubated at 25°C for 24 h. The unsettled microalgae on the coating surface were rinsed off with artificial seawater before the characterization of the attached microalgae. All quantitative measurements were carried out on at least three samples per group.

Marine Field Tests: The actual sea trials were carried out during June to August 2020 at the Xiangshan harbor ($29^\circ49'\text{N}$, $122^\circ0'\text{E}$) in China according to the China Standard (GB/T 5370-2007). During this period, the mean seawater temperature was around 27 to 28°C and water pH was around 8 ± 0.1 . The commercially purchased coating resin without antifouling agent was added with 1%, 5%, and 10% (w/w) *E*-DFFD-Cu, and then dispersed in a multifunctional mixer at 3000 rpm for 60 min, and then ground in a sand mill to a fineness of 20 μm and filtered through a 200 mesh stainless steel filter to obtain the desired antifouling paint. The prepared antifouling paint was blended with diluent to a suitable viscosity and sprayed on the surface of the anticorrosion coated sample at 0.3–0.4 MPa with the film thickness controlled at $(100 \pm 2) \mu\text{m}$. The coated sample was maintained in a constant temperature chamber at 25°C for 7 days and then used for the real sea hanging test. Coated plates were mounted on stainless steel panels (190 mm $\times$ 70 mm $\times$ 1 mm) that were statically immersed into the sea in 0.2–2.0 m dept. The panels were removed from the seawater and photographed once a month.

Membrane Permeability Tests: 100 μL bacterial suspension with 10^6 CFU mL^{-1} was placed in a 24-well plate. *E*-DFFD and *E*-DFFD-Cu were then added into the bacterial suspension, respectively. After 2 hours of treatment at 37°C , 1000 μL samples were collected, centrifuged and washed with phosphate buffered saline (PBS) solution for three times. 200 μL of dye propidium iodide (PI) solution with a concentration of 19.80 $\mu\text{mol L}^{-1}$ was then added to above samples and stained for 30 min at room temperature. The samples were centrifuged at 4000 rpm to remove the supernatant and washed with PBS solution for three times. 50 μL samples were collected and imaged by fluorescent inverted microscope (SP5, LEICA).

Morphological Characterization: The prepared bacterial solutions of *E. coli* and *V. parahaemolyticus* were treated and incubated with *E*-DFFD-Cu of 4.95 $\mu\text{mol L}^{-1}$ at 37°C mixing at 50 rpm for 4 and 24 h, respectively. And then bacterial cells were collected by centrifugation at

8000 rpm for 5 min at 4 °C. The precipitated bacterial cells were immediately fixed in 2.5% glutaraldehyde solution at 4 °C for 10 h, and then washed three times with PBS solution followed by serial dehydration with 30%, 50%, 70%, 90%, and 100% ethanol solutions at 15 min intervals. After freeze-drying for 12 h, the cells were covered coated with gold by means of sputter coater (BALTEC SDC 050 Sputter Coater) and then observed under a ZEISS EVO18 scanning electron microscope.

Radical Measurements: Radicals were measured by electron-paramagnetic resonance (EPR) technology with DMPO as spin-trap reagent. During a typical run, the E-DFFD and E-DFFD-Cu compounds were dissolved in phosphate-buffered solution (PBS) to a concentration of 4.95 $\mu\text{mol L}^{-1}$. Then DMPO was added to the samples with a concentration of 1.0 mmol L^{-1} . The mixture was then immediately transferred into the EPR quartz flat cell and analyzed by EPR. EPR spectra were recorded at room temperature on Bruker E500 EPR spectrometer applying the following instrumental settings: microwave power 0.63 mW, scanning frequency 9.8 GHz, central field 3510 G, scanning width 200 G, modulation frequency 100 kHz, modulation amplitude 1.0 G, conversion time 40 s, number of scans 6.

Determination of the Release Rates of Copper Ion and E-DFFD: The antifouling duration of each coating was estimated based on the release rate of E-DFFD and E-DFFD-Cu from antifouling coating samples in artificial seawater through measuring their concentrations according to the ISO 15181-2007 standard. It comprises immersion of brush-painted resins substrates (10 mm $\times$ 10 mm $\times$ 1 mm) in artificial seawater (20 mL) under static condition and for a minimum period of 30 days. The temperature was maintained at 25 °C. The immersed samples were tested for each type of E-DFFD and E-DFFD-Cu composite coatings to obtain average values.

The release rate of E-DFFD and E-DFFD-Cu from the coating samples was further extrapolated based on their cumulative concentrations (Equations (S1–S3)) (Section S2.8, Supporting Information). Three samples were tested for each type of E-DFFD and E-DFFD-Cu composite coatings to obtain average values.

Bio-Toxicity Tests: The acute toxicity tests with the marine bacteria *Vibrio fischeri* was performed according to the GB/T 15441-1995 standard method (GB/T 15441, 1995). This test was used to determine toxicity based on the value of EC_{50} . Toxicity tests were conducted under controlled temperatures at 15 °C $\pm$ 1 °C for exposure and 3 °C $\pm$ 3 °C for the bacterial reagent. The EC_{50} values were calculated using a Microtox Omni v. 4.0 software (AZUR Environmental-Carlsbad, USA).

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

Keywords

5-diformylfuran dioxime, anti-biofoulant, antifouling coating, bio-based 2, copper(II) catalysis, low toxicity

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